

Bayesian Hierarchical Model for PM2.5 in Italy

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1 Introduction

Air pollution represents a major environmental and public health concern. In particular, fine particulate matter (PM2.5) is among the most harmful pollutants due to its ability to penetrate deep into the respiratory system and its strong association with cardiovascular and respiratory diseases.

A key challenge in modeling PM2.5 concentrations lies in the presence of substantial heterogeneity across monitoring stations and irregular temporal coverage. Stations differ in terms of geographical characteristics, local emission sources, and environmental conditions, while observational data often exhibit an unbalanced panel structure, with varying numbers of observations across stations and missing time periods.

In this context, Bayesian hierarchical models provide a natural and flexible framework. By introducing group-level random effects, these models allow for partial pooling, enabling information sharing

across stations while preserving local heterogeneity. Moreover, the Bayesian approach provides a coherent framework for uncertainty quantification and prior specification.

This paper develops a Bayesian hierarchical model for daily PM2.5 concentrations observed across monitoring stations in Italy. Posterior inference is carried out using a Metropolis-within-Gibbs sampler. The analysis focuses on model specification, MCMC implementation, convergence diagnostics, sensitivity analysis, and posterior predictive checks.

2 Data

The dataset consists of daily PM2.5 measurements collected across monitoring stations in Italy during 2025. After preprocessing, the final dataset contains 31,349 observations across 116 stations.

The empirical distribution of the raw outcome variable is highly right-skewed, with a skewness of approximately 17. This strong asymmetry is driven by extreme values in the upper tail. To address this issue, a logarithmic transformation is applied:

$$\tilde{y}_i = \log(y_i),$$

which substantially reduces skewness (approximately -0.29) and yields a more symmetric distribution, supporting the use of a Gaussian likelihood. Figure 1 illustrates the strong right-skewness of the raw PM2.5 concentrations and the substantial improvement in symmetry obtained after the logarithmic transformation.

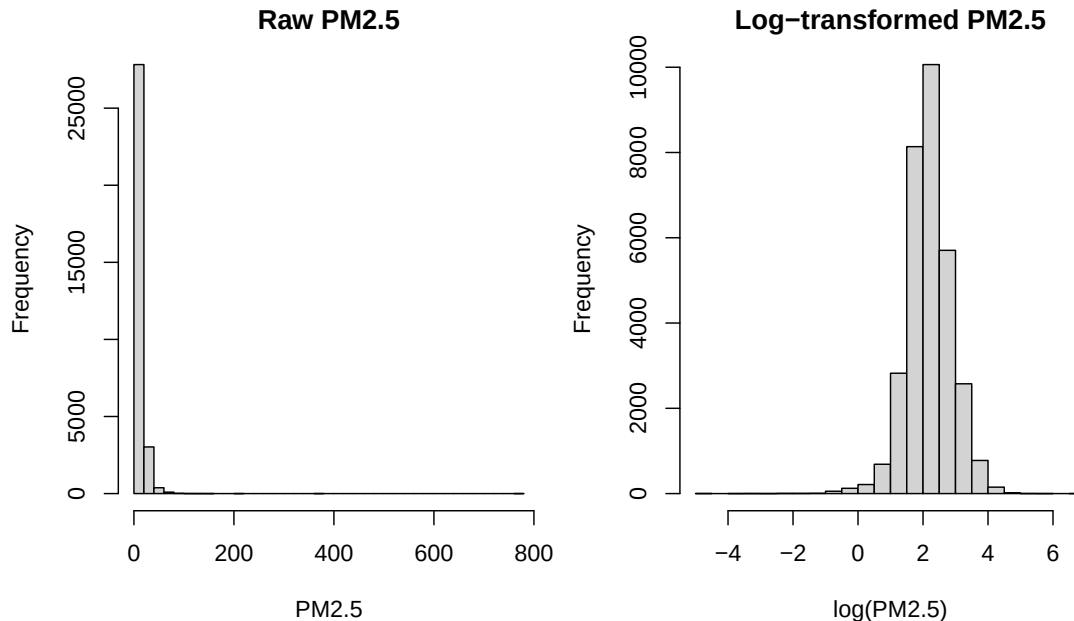


Figure 1: Distribution of raw and log-transformed PM2.5 concentrations.

The panel structure is markedly unbalanced. The number of observations per station ranges from 69 to 332, with a median of approximately 299. Additionally, temporal coverage varies across stations: some are observed for the entire year, while others are only observed for a limited number of months.

In particular, 32 stations exhibit at least one completely missing month, and the minimum coverage is only 4 months. The unbalanced panel structure is clearly visible in Figure 2.

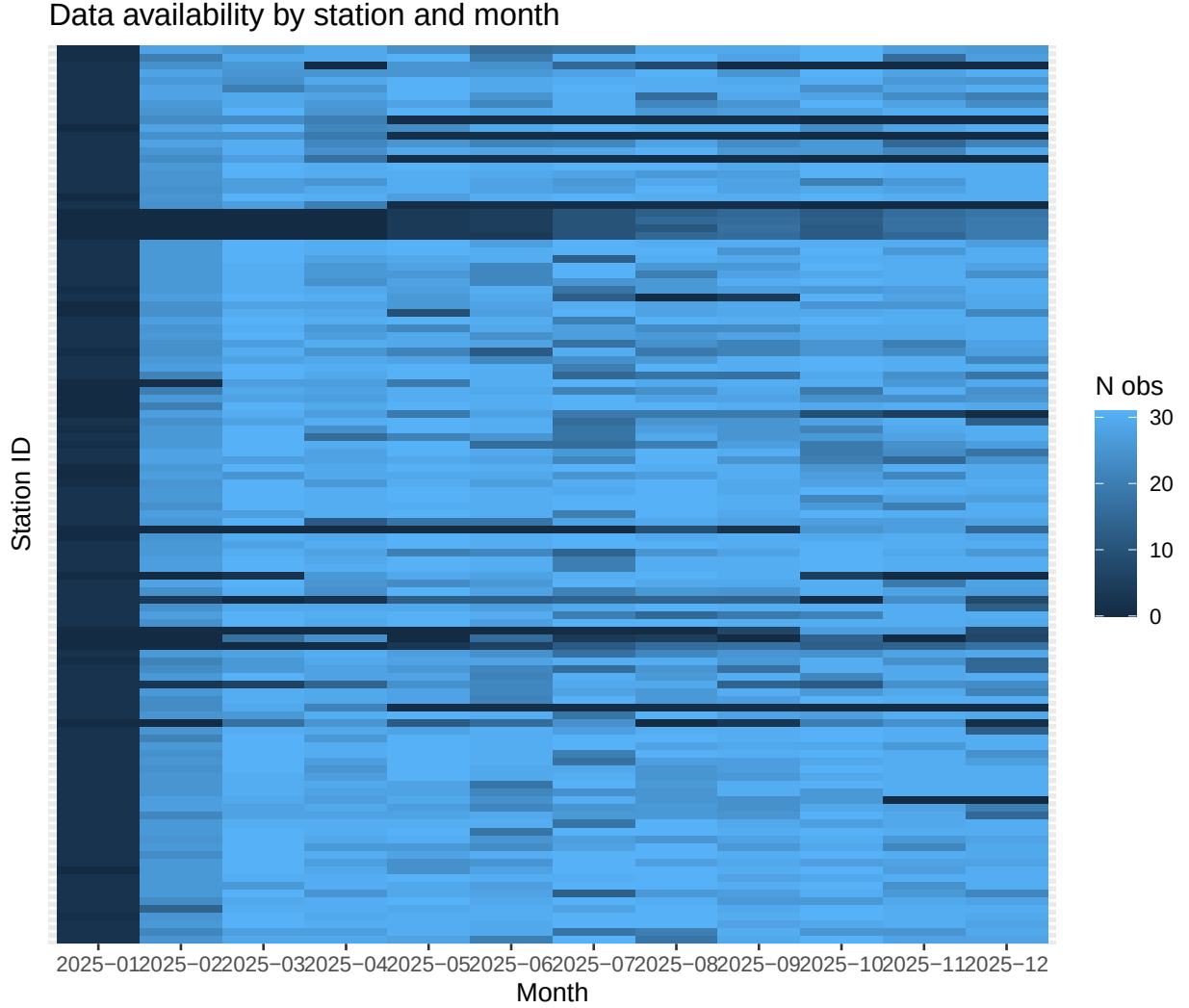


Figure 2: Data availability by station and month. The panel is highly unbalanced, with heterogeneous temporal coverage and several missing station-month combinations.

These features motivate the use of a hierarchical specification, which is well suited to handle unbalanced grouped data.

3 Model

We consider a hierarchical model for the log-transformed PM2.5 concentrations. Let $\tilde{y}_i$ denote the transformed outcome for observation i . The likelihood is specified as:

$$\tilde{y}_i \mid \beta, \alpha, \sigma^2 \sim \mathcal{N}(x_i^\top \beta + \alpha_{j[i]}, \sigma^2),$$

where: - x_i is a vector of covariates including month and day-of-week effects, - β is a vector of

regression coefficients, - $\alpha_{j[i]}$ is a station-specific random intercept.

3.1 Hierarchical structure

The random effects are modeled as:

$$\alpha_j \mid \tau \sim \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J,$$

which induces partial pooling across stations.

3.2 Prior specification

Weakly informative priors are assigned to all parameters.

For the regression coefficients:

$$\beta \sim \mathcal{N}(b_0, B_0),$$

where $B_0 = cI$ is a diagonal covariance matrix.

For the observation variance:

$$\sigma^2 \sim \text{Inverse-Gamma}(a_\sigma, b_\sigma).$$

For the group-level standard deviation:

$$\tau \sim \text{Half-Cauchy}(0, A).$$

The use of a half-Cauchy prior for τ avoids the well-known sensitivity of inverse-gamma priors for variance components, while providing weakly informative regularization.

3.3 Posterior inference

Posterior inference is performed using a Metropolis-within-Gibbs sampler. The full conditional distributions for β , α_j , and σ^2 belong to standard families and are sampled via Gibbs steps.

In contrast, the full conditional distribution of τ is not available in closed form due to the half-Cauchy prior. Therefore, a Metropolis-Hastings step is used, implemented on the logarithmic scale with a normal random-walk proposal.

The sampling scheme can be summarized as follows:

- Sample β from its conditional posterior
- Sample each α_j from its conditional posterior
- Sample σ^2 from its conditional posterior
- Update τ via a Metropolis-Hastings step

4 MCMC Implementation

The sampler is run for 5,000 iterations. Visual inspection of the trace plots suggests a short initial transient phase, and the first 200 iterations are discarded as burn-in. To reduce autocorrelation, a thinning scheme is applied by retaining every 5th draw.

4.1 Tuning of the Metropolis step

The proposal distribution for τ is a normal random walk on the log scale. A tuning exercise was conducted to assess the impact of the proposal standard deviation.

With a proposal standard deviation of 0.15, the acceptance rate was approximately 0.48. Increasing the proposal scale to 0.20 reduced the acceptance rate to approximately 0.38. Despite this change, trace plots and autocorrelation functions remained essentially unchanged, and posterior summaries were virtually identical.

Given these results, the value `proposal_sd = 0.20` is retained in the final implementation.

5 Results

5.1 Convergence diagnostics

Trace plots of the post-burn-in chains are shown in Figure 3. There is no evidence of non-stationarity or systematic trends. The chains fluctuate around stable levels, indicating that convergence has been achieved.

Residual autocorrelation is illustrated in Figure 4, motivating the use of thinning in posterior summaries.

The Geweke diagnostic confirms this result. All z-scores are well within the conventional acceptance range, providing no evidence against convergence.

The effective sample size (ESS) is generally high across parameters. In particular, the variance parameter exhibits an ESS close to the total number of retained draws, while the hyperparameter τ also shows a high ESS. The random effects display more variability, as expected, but remain within acceptable ranges.

5.2 Posterior summaries

Table 1 reports the posterior means of the key model parameters.

The posterior mean of the intercept is approximately 2.60 (on the log scale), corresponding to an average PM2.5 level of about $\exp(2.60) \approx 13.5$.

The residual variance σ^2 is estimated around 0.32, implying a standard deviation of approximately 0.56. This reflects the within-station variability in daily PM2.5 concentrations.

The group-level standard deviation τ is approximately 0.30, capturing the variability across monitoring stations.

Comparing σ^2 and τ suggests that within-station variability is larger than between-station variability, indicating that short-term fluctuations dominate over persistent spatial differences, although the latter remain non-negligible.

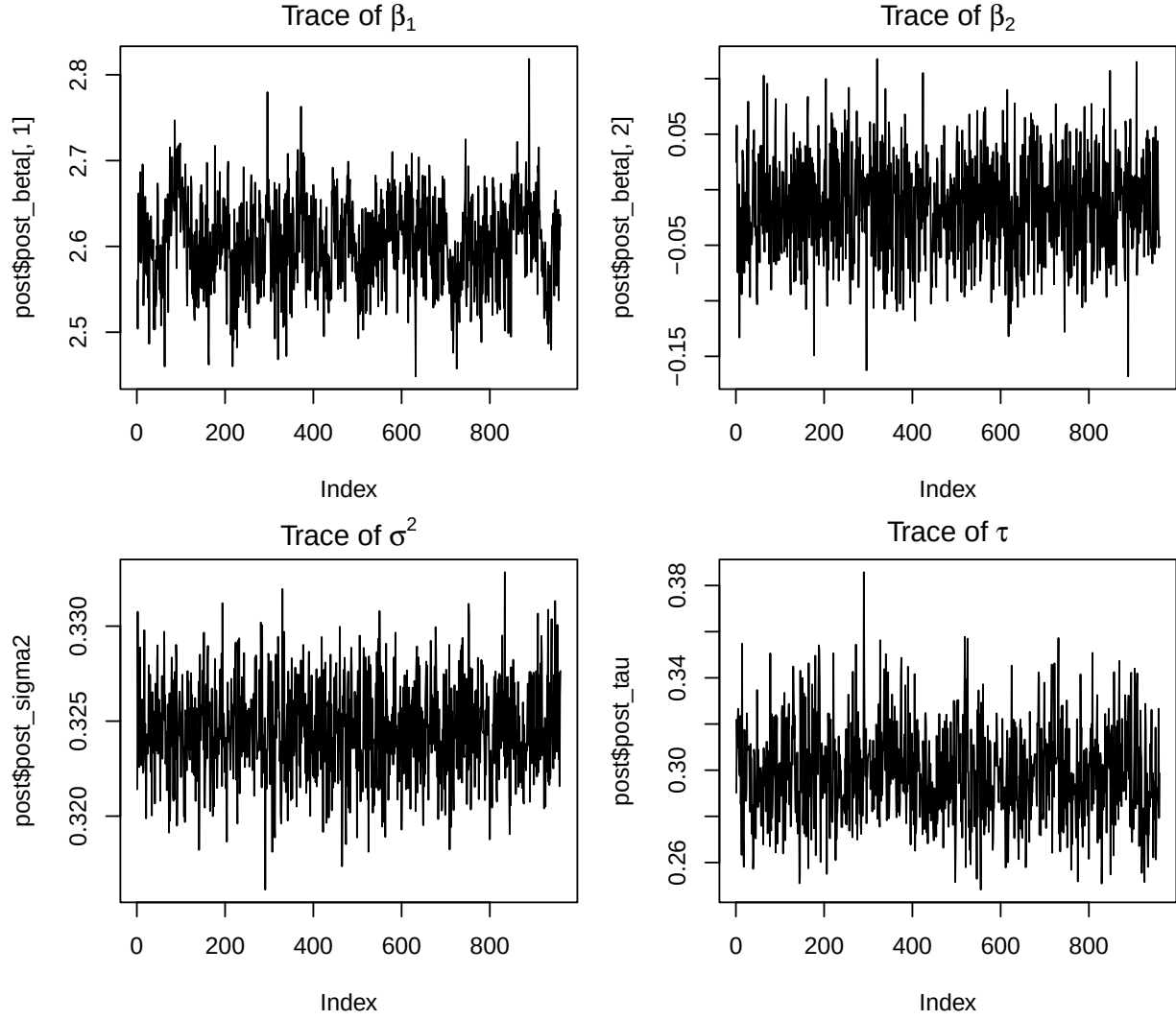


Figure 3: Post-burn-in trace plots for selected model parameters. The chains fluctuate around stable levels without visible trends, supporting convergence of the Metropolis-within-Gibbs sampler.

Table 1: Posterior summary of key model parameters.

Parameter	Mean
Intercept (β_0)	2.60
Residual variance (σ^2)	0.32
Group-level sd (τ)	0.30

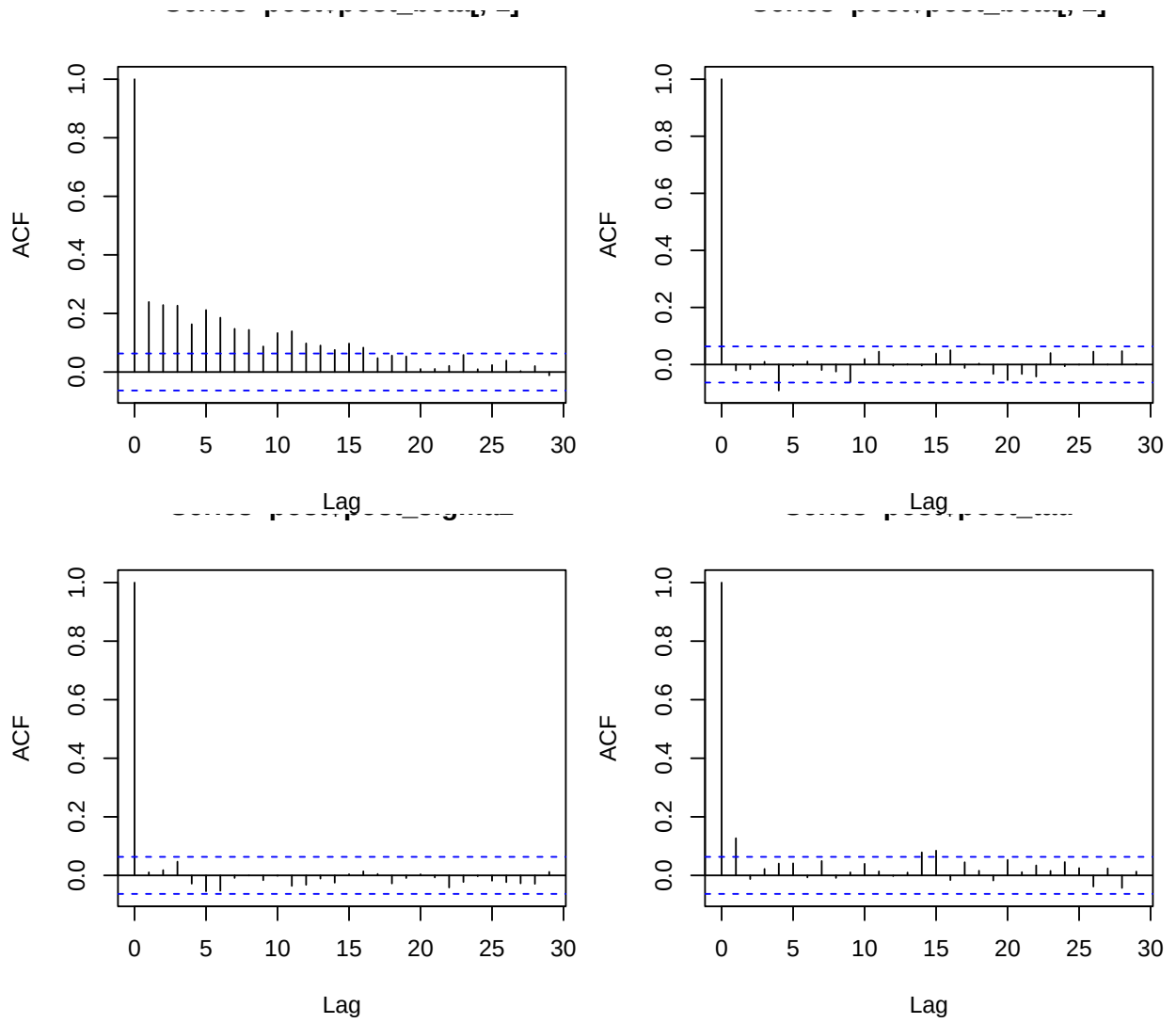


Figure 4: Post-burn-in autocorrelation functions for selected model parameters. Residual autocorrelation is limited for most parameters, although some mild dependence persists.

5.3 Sensitivity analysis

A sensitivity analysis was conducted by varying key prior hyperparameters, including: - the prior variance of β , - the scale parameter of the half-Cauchy prior on τ , - the parameters of the inverse-gamma prior on σ^2 .

The results show that posterior estimates are remarkably stable across all specifications. In particular, the posterior means of the intercept, σ^2 , and τ remain virtually unchanged. This suggests that the data are highly informative and that the posterior distribution is largely driven by the likelihood.

5.4 Posterior predictive checks

Posterior predictive checks were performed to assess model adequacy. Replicated datasets were generated from the posterior distribution and compared to the observed data.

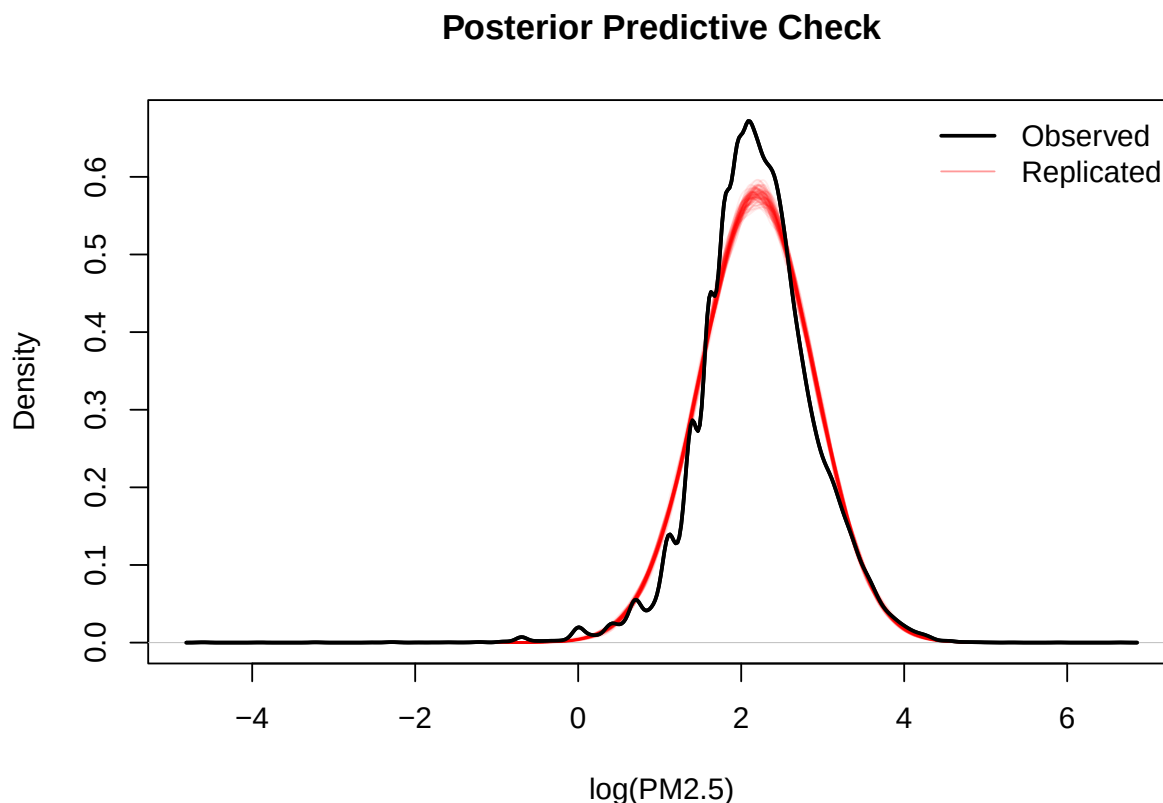


Figure 5: Posterior predictive check for the log-transformed PM2.5 outcome.

The distribution of replicated means and standard deviations is consistent with the observed values (see Figures @ref(fig:fig-ppc-density and 6, indicating that the model accurately captures the first two moments of the data.

Furthermore, the posterior predictive density aligns well with the empirical distribution of log-transformed PM2.5, with only minor discrepancies in the peak and tail regions. These differences

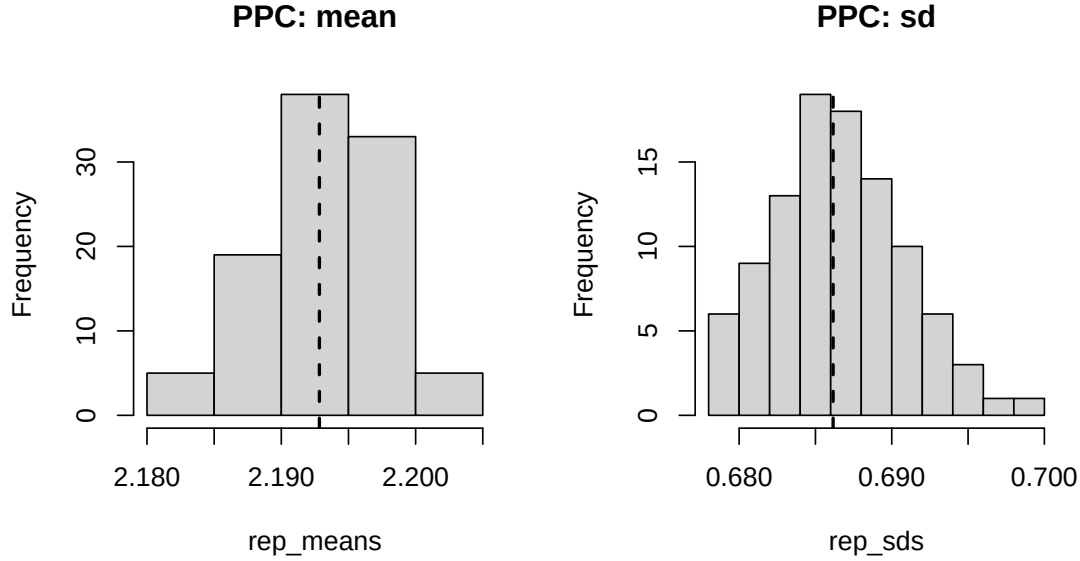


Figure 6: Posterior predictive checks based on summary statistics. The observed mean and standard deviation fall well within the distribution of replicated values.

are consistent with the Gaussian assumption and do not indicate major model misspecification.

5.4.1 Interpretation of results

The posterior estimates provide a clear interpretation of the main sources of variability in PM2.5 concentrations.

The posterior mean of the intercept is approximately 2.60 on the log scale. This corresponds to an average PM2.5 level of about $\exp(2.60) \approx 13.5$, which can be interpreted as a baseline pollution level after accounting for temporal effects. This value is consistent with typical moderate air pollution conditions.

The residual variance σ^2 is estimated to be around 0.32, implying a standard deviation of approximately 0.56 on the log scale. This component captures the day-to-day variability in PM2.5 concentrations within each monitoring station, reflecting short-term fluctuations due to meteorological conditions and local emission patterns.

The group-level standard deviation τ is estimated at approximately 0.30. This parameter measures the variability across monitoring stations, capturing persistent differences in pollution levels due to geographical, environmental, and structural factors.

Comparing these two sources of variability provides an important insight: the within-station variability is larger than the between-station variability. This suggests that short-term fluctuations in air pollution are more pronounced than systematic differences across locations. However, the magnitude of τ still indicates a non-negligible degree of heterogeneity across stations, justifying the hierarchical specification.

Overall, these results highlight that both temporal variability and spatial heterogeneity play a role

in shaping PM2.5 dynamics, with the hierarchical model effectively capturing both components.

6 Conclusion

This paper develops a Bayesian hierarchical model for daily PM2.5 concentrations, accounting for heterogeneity across monitoring stations and unbalanced panel structure.

The Metropolis-within-Gibbs sampler exhibits good convergence properties, supported by multiple diagnostic tools. Posterior inference is robust to prior specification, and posterior predictive checks indicate a satisfactory model fit.

Overall, the hierarchical Bayesian framework provides a flexible and effective approach for modeling air pollution data. Possible extensions include the incorporation of spatial dependence across stations, the use of more flexible likelihood specifications, and the inclusion of additional covariates capturing environmental and meteorological factors.